SHINEY RASHMITHA CHANDRAGANTI

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EDUCATION

Master of Science, Biotechnology and Entrepreneurship Sep 2021 – May 2023 | New York City, USA
New York University, Tandon School of Engineering
Coursework: Cancer Immunology, Health Care, Biosensors & Biocatalysis

Bachelor of Technology, Biotechnology Jul 2016 – Dec 2020 | Allahabad, India
Motilal Nehru National Institute of Technology
Coursework: Immunology, Genetics, Cell and System Biology, Molecular Biology & Biostatistics

RESEARCH EXPERIENCE

Cornell University, Dept. of Biomedical Sciences
Lab Technician III Aug 2023 – present | Ithaca, NY, USA
Research Experience
Advisor: Dr.Anushka Dongre, PhD, Department of Biomedical Sciences
Project: *CD4+ T-cells sensitize quasi-mesenchymal breast tumors lacking CD73 to anti-CTLA4 immune checkpoint blockade therapy*

- Conducted systematic depletion and complementary knockout models (CD4-KO, CD8-KO, MHC-II-KO) to identify which T-cell population mediates anti-CTLA4 response in CD73-deficient breast cancer tumors
- Generated CD73/B2M double knockout tumor cell lines using CRISPR/Cas9 to exclude MHC-I dependency and validated by Western blot and flow cytometry
- Executed multi-parameter (14+ colors) flow cytometry to phenotype T-cell subsets to define CD4+ T-helper polarization in non-responding and responding tumors
- Carried out orthotopic tumor implantation and rechallenge studies in immunocompetent mice to assess therapeutic efficacy and memory formation against breast cancer tumors
- Performed immunofluorescence staining to quantify CD4+/CD8+ T cell infiltration and validate response mechanisms through microscopic analysis
- Compiled research findings and drafted manuscript submitted to Cancer Research Communications, currently under review

Techniques: CRISPR/Cas9 gene editing, Cell Culture, Orthotopic tumor implantation and IP injections, Flow Cytometry, Immunofluorescence Staining, Confocal Microscopy, and Statistical Analysis (GraphPad Prism)

Advisors: Dr.Anushka Dongre, PhD, and Dr.Glenn Jr. Simmons, Department of Biomedical Sciences
Project: *Understanding the role of lipid metabolism in epithelial-mesenchymal transition in breast cancers*

- Conducted Western blot to investigate whether SCD1 knockdown in epithelial cells induces epithelial-mesenchymal plasticity (EMP) by looking at expression of EMT markers
- Carried out Cytoplasmic Immunofluorescence to verify induction of epithelial to mesenchymal cell phenotype when SCD1 is knocked down in epithelial cells by looking into various morphological changes
- Performed orthotopic mammary fat pad implantations of GFP-labeled epithelial, mesenchymal, and SCD1-knockdown breast cancer cell lines in immune-competent syngeneic mice to establish primary tumors

Techniques: Western blot, Cytoplasmic immunofluorescence, Orthotopic mammary fat pad implantation

Lab Management

- Managed mouse colonies across 6 breeder lines, including daily inspections, breeder pair setup, weaning, and experimental organization

- Optimized genotyping protocols, DNA extraction, PCR analysis, and gel electrophoresis for enhanced colony management
- Implemented lab management systems, including MS Excel-based inventory tracking, weekly supply check-in/out protocols, and detailed experimental records to reduce time conflicts and duplicate ordering
- Operated and maintained laboratory equipment such as microscopes and centrifuges while performing routine maintenance tasks, including liquid nitrogen and CO2 level management, glassware sterilization, and safe disposal of hazardous chemical and biological waste following safety protocols
- Trained 7 graduate and 6 undergraduate researchers in tissue culture, immunostaining, microscopy, and flow cytometry analyses
- Established collaborative relationships with other research laboratories and core facilities, delivering interactive presentations and posters to showcase research findings

NYU School of Global Public Health
Graduate Research Assistant Sep 2022 – Jun 2023 | New York City, USA
Advisor: Dr.Andrea L Deierlein, PhD, Director of Public Health Nutrition
- Recruited 210+ participants into study, conducted protocol briefings, and obtained informed consent in compliance with ethical guidelines
- Handled distribution of glucose monitoring devices and implemented flexible travel schedules to accommodate participants, maintaining high participation and proper data collection protocols for dietary analysis
- Analyzed dietary data across three time points per participant to evaluate correlations between nutritional, behavioral, and environmental factors on maternal health outcomes

Techniques: Participant recruitment and informed consent, distribution and scheduling, data collection, and correlation analysis of health factors

NYU Langone Health, Alexandria Center for Life Science
Graduate Research Student Jul 2022 – May 2023 | New York City, USA
Advisor: Dr.Kamal M. Khanna, PhD, Department of Microbiology
Project: *Spatial characterization of Tumor-Associated Macrophages (TAMs) in lung tumor microenvironment*
- PCR genotyping of multi-transgenic mice and maintained various cell lines for downstream *in vivo*-based research studies investigating innate and adaptive immune system dynamics in lung tumors
- Induced murine bone marrow-derived macrophages to evaluate impact of certain gene products on macrophage function and immune responses
- Executed Cryosectioning and multiplex immunofluorescence staining of murine tissue samples to visualize using a Keyence BZ-X800 epifluorescence and/or a Zeiss LSM 880 confocal microscope
- Applied ImageJ and Imaris image analysis software for quantitative analysis of immune cell populations and interaction dynamics

Techniques: PCR genotyping and cell culture, Macrophage induction, Cryosectioning and multiplex Immunofluorescence Staining, Fluorescence and Confocal microscopy, and Image analysis (ImageJ, Imaris)

NYU Langone Health, Institute for Computational Medicine
Graduate Research Volunteer Feb 2022 – Aug 2022 | New York City, USA
Advisor: Dr.Jonas Schluter, PhD, Department of Microbiology
- Investigated research gaps in quorum sensing communication systems in gut bacteria through literature review and computational analysis
- Extracted microbial DNA from a lymphoma patient sample using QIAcube and quantified DNA via spectrophotometer to evaluate intestinal microbiome impact on immune functions

Techniques: Computational analysis, Microbial DNA extraction (QIAcube), and DNA quantification (spectrophotometry)

MNNIT Allahabad, Department of Biotechnology
Undergraduate Research Student Jan 2020 – Dec 2020 | Allahabad, India
Advisor: Dr.Shivesh Sharma, PhD, Department of Biotechnology
Collaborator: Jaspreet Singh, PhD Student
Project: *Computational studies to identify potential inhibitors in Type II Secretion System in pathogenic bacteria*
- Conducted high-throughput virtual screening of chemical libraries to identify potential inhibitors for Type II Secretion System in pathogenic Gram-negative bacteria including *P. aeruginosa and E. coli*
- Performed molecular docking analysis to evaluate binding affinities and interaction patterns between potential inhibitors and target proteins
- Assessed pharmacokinetic properties, toxicity profiles, and ADME (Absorption, Distribution, Metabolism, and Excretion) characteristics using computational tools
- Compiled all research findings and contributed to thesis documentation for my senior year project

Techniques: High-throughput virtual screening, Molecular docking, Pharmacokinetic analysis, ADME profiling, and toxicity analysis

IIT Guwahati, Department of Biosciences and Bioengineering
Undergraduate Research Intern May 2019 – Jul 2019 | Guwahati, India
Advisor: Dr.Lalit M. Pandey, PhD, Department of Biosciences and Bioengineering
Project: *Synthesis and characterization of methyl acrylamide cellulose nanowhiskers for environmental applications*
- Synthesized and modified cellulose nanowhiskers (CNWs) chemically through acid hydrolysis and surface functionalization to enhance biocompatibility and adsorption capacity
- Performed extensive literature review and optimized experimental protocols for methyl acrylamide modification to maximize functionalization efficiency
- Contributed to experimental design, data visualization and results interpretation, and manuscript writing, resulting in publication in Letters of Applied NanoBioScience

Techniques: Acid hydrolysis, Functional and Chemical Characterization, Data analysis, and Scientific writing

Aurobindo Pharma Ltd., Department of Microbiology and Molecular Biology
Pharmaceutical Intern May 2018 – Jul 2018 | Hyderabad, India
- Observed manufacturing processes from bench-scale to final commercial product in therapeutic drug industry, understanding regulatory compliance and quality assurance protocols
- Applied fundamental techniques including PCR, DNA isolation, molecular cloning with plasmid vectors, and protein extraction using SDS-PAGE for quality control testing
- Gained exposure to translational research demonstrating how biomaterials can be engineered to market-ready therapeutic products

Techniques: PCR, Gel electrophoresis, DNA isolation, Molecular Cloning, Quality control, and GMP compliance

MNNIT Allahabad, Department of Biotechnology
Undergraduate Research Student Aug 2016 – May 2017 | Allahabad, India
Advisor: Radha Rani Mewaram, PhD, Department of Biotechnology
Project: *Effect of different substrates as an electron donor on the performance of Microbial Fuel Cell (MFC)*
- Investigated microbial fuel cell (MFC) performance using *P. gessardii* as biocatalyst for sustainable energy generation
- Manipulated environmental factors including glucose concentrations and substrate types to optimize MFC performance and efficiency
- Evaluated potential applications of biological systems for sustainable energy production and environmental remediation

Techniques: Microbial fuel cell operation, Culture optimization, Electrochemical measurements, and Metabolic analysis

HONORS & AWARDS

- MNNIT Allahabad Admission - India's 3rd-ranked NIT (Top 0.1% of 1.04M candidates) 2016
- Board of Intermediate Education - 3.9 GPA Maths, Physics, Chemistry 2016
- Secondary School Excellence - 3.9 GPA 2014
- International Mathematics Olympiad Participant 2010, 2011, 2013
- National Science Olympiad Participant 2011, 2013
- Mathematics Olympiad Award - City Rank 38, State Rank 1,026 2010
- State Level Science Talent Search Examination - Top 1,000 students 2009

PUBLICATIONS

1. **Shiney Chandraganti**, Caitlyn Sams, Sarthak Sahoo, Brian Feng, Isabel O'Connell, Lynna Li, Sunita Nepal, Siddhartha Pulukuri, Kimaya Bakhle, Prathapan Thiru, Corina Simian, George W. Bell, Mohit Kumar Jolly, Anushka Dongre (2025). "CD4+ T-cells sensitize quasi-mesenchymal breast tumors lacking CD73 to anti-CTLA4 immune checkpoint blockade therapy." (Under review at Cancer Research Communications) **bioRxiv preprint** doi: https://doi.org/10.1101/2025.05.12.653467
2. Eric Bartnicki; Kamal Mohan Khanna; Eduardo Brenier; Romane Cathelin; Stephen Yeung; Payal Damani-Yokota; Mimi Kostoska; Dhar Vinica; Dhairya Upadhyay; Ray Pillay; **Shiney Chandraganti**; Thales Papagiannakopoulos; Nathalie Pinkerton; Harvey Pass; Kwok-Kin Wong; Bettina Nadorp; Aristotelis Tsirigos. "Cross-species conservation of nerve- and airway-associated interstitial macrophages (NAMs) that suppress $CD8^+$ T cells immunity via a C1q axis in lung cancer" (Under review at Immunity, Cell Press)
3. Swati Sharma, **Ch. Shiney Rashmitha** and Lalit M. Pandey (2020). "Synthesis and characterization of methyl acrylamide cellulose nanowhiskers for environmental applications." doi: https://doi.org/10.33263/LIANBS91.880884
4. Kimaya Bakhle, **Shiney Chandraganti**, Sophie Nelissen, Lynna Li, Brian Feng, Brenda Ramos-Villanueva, Andrew Miller and Anushka Dongre. "The epithelial-mesenchymal transition drives CD73-mediated cross-protection in heterogeneous breast tumors" (In preparation)
5. Brian Feng, Caitlyn Sams, Isabel O'Connell, **Shiney Chandraganti**, Sarthak Sahoo, Mohit Kumar Jolly, Anushka Dongre. "Understanding the mechanism by which CD4+ T cells sensitize quasi-mesenchymal sgCD73 tumors to anti-CTLA4 checkpoint blockade therapy" (In preparation)
6. Sophie Nelissen, Laila Salih, **Shiney Chandraganti**, Isabel O'Connell and Anushka Dongre. "Harnessing epithelial-mesenchymal plasticity (EMP) in breast cancer to potentiate the effects of immunotherapy" (In preparation)

ABSTRACTS

"CD4+ T-cells sensitize quasi-mesenchymal breast tumors lacking CD73 to anti-CTLA4 immune checkpoint blockade therapy." **Intercampus Cancer Metabolism Symposium**, Cornell University, Ithaca, NY, 2025

"CD4+ T-cells sensitize quasi-mesenchymal breast tumors lacking CD73 to anti-CTLA4 immune checkpoint blockade therapy." **24th Annual Biomedical & Biological Sciences Symposium**, Department of Biomedical Sciences, Cornell University, Ithaca, NY, 2025

"Understanding the role of CD4+ T cells in sensitizing mesenchymal tumors to immune checkpoint blockade therapy." **Biomedical & Biological Science (BBS) Recruitment Poster Session**, Cornell University, Ithaca, NY, 2025

"Sensitizing quasi-mesenchymal breast tumors to immune checkpoint blockade therapy by targeting CD73" **International Conference - TEMTIA 11,** Seattle, WA, 2024

"Understanding the role of CD4+ T cells in sensitizing mesenchymal tumors to immune checkpoint blockade therapy." **Genetics, Genomics, and Development-Biochemistry and Structural Biology, Cell & Developmental Biology, and Molecular & Cell Biology (GGD-BMCB) Recruitment Poster Session**, Cornell University, Ithaca, NY, 2024

"Understanding the role of CD4+ T cells in sensitizing mesenchymal tumors to immune checkpoint blockade therapy." **Biomedical & Biological Science (BBS) Recruitment Poster Session**, Cornell University, Ithaca, NY, 2024

"Sensitizing quasi-mesenchymal breast tumors to immune checkpoint blockade therapy by targeting CD73." **Biomedical & Biological Science (BBS) Recruitment Poster Session**, Cornell University, Ithaca, NY, 2023

PROFESSIONAL DEVELOPMENT

- **Graduate Coursework in Veterinary Medicine** Fall 2023

Cornell University - *VTBMS 7010: Mouse and Stem Cell Pathology*
- Completed advanced-level coursework in pathology principles and methods while employed as a Lab Technician
- Gained proficiency in histotechnology techniques, mouse necropsy procedures, and systematic pathological evaluation

- **Graduate Coursework in Biomedical Sciences** Fall 2024

Cornell University - *BIOMS 6150: Essential Immunology*
- Completed advanced graduate-level coursework in immune system fundamentals and immunological concepts while employed as a Research Technician
- Gained expertise in cell biology of immune signaling, genetic recombination, and lymphocyte development
- Applied knowledge of vaccine mechanisms, immunotherapies, and immune-mediated diseases to enhance understanding of current laboratory projects
- Presented a poster on "**Evidence for Original Antigenic Sin of Omicron SARS-CoV-2 Immune Response"** in Biomedical Sciences Department

MENTORSHIP

Undergraduate Students

- Siddhartha Pulukuri (Microbiology and Immunology) 2024-present
- Kaden Priebe (Biological Sciences) 2024-present
- Laila Salih (Biological Sciences, McNair Scholar) 2025-present
- Caitie Sams (Biological Sciences and Chemistry, Hunter Rawlings III Presidential Scholar) 2023-2025
- Ashley Teal (Nutritional Sciences) 2023-2025

High School Students

- Aanya Jain Summer 2025

LEADERSHIP

Student Assembly

Team Leader, AGORIZE: Bayer Digital Campus Innovation Challenge 2021
Led a multidisciplinary team of 4 in a global pharmaceutical innovation competition among 1100+ participants from 82 countries, coordinated project development and strategy for a digital healthcare solution addressing 'Good Digital Health Practice'
Public Affairs Team Member, CULRAV Cultural Fest & AAVISHKAR Tech Fest, NIT Allahabad 2017-2019
Organized different cultural and technical festivals engaging diverse student populations across campus
Event Coordinator, Biotechnology Department, NIT Allahabad 2017-2019
Participated in organizing departmental events, including Central Dogma, Genesis, and Palladin, promoting academic community engagement

Co-curriculars & Outreach Experience

Active Participant, Bridges International NYU 2021-2023
Engaged in peer support events throughout every semester for international students, contributed to community-building, and fostered cross-cultural understanding
Telugu Students Community - GULTY, NIT Allahabad 2016-2020
Coordinated cultural and festival events connecting 150+ Telugu-speaking community members within a diverse campus environment
Cultural Performer, GARBA Dance Group, NIT Allahabad 2016
Represented traditional Gujarati dance forms at university cultural fest, promoting cultural diversity within campus community
District Sports Champion 2011
Triple gold medalist in 100m Sprint, 200m Sprint, and Kho-Kho events, which show mastery in both team and individual sports
1st Prize Winner in Drawing Competition, 51st Annual Celebration, LIC of India 2011
Drawing Competition Winner in 51st Insurance Week Celebrations sponsored by Life Insurance Corporation of India, which shows artistic and creative skills